ECON 100B Spring 2026 Your name and SID: Marcus Huang 3041432436

Problem Set 3

Due on Gradescope: check the course Drive calendar
Consult the Late Policy in the syllabus

Instructions: Work with this file as a template, perhaps by File: Download: Microsoft Word. Other options, like handwriting, are fine as long as they are legible. Name and SID at the top.

Did you work with other students? Identify them below. **If you work with others, you should write answers in your own words.** (If you understand your answers, this should be easy.)

Type your name to sign the Honor Code. Save as a PDF, upload to Gradescope, and click through to locate each question part before clicking Submit. Profit.

You will see point totals associated with each question. But we will score your problem sets based on completion, not correctness. The point totals are there to make you familiar with how the exams will work, which are scored on correctness.

Questions below:

Did you work with other students? List them below:

No, I am working on my own.

Please type your name as an affirmation of the Honor Code at the University of California, Berkeley.

> "As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others."

Type your name:

Marcus Huang

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1. [6 points total] Labor markets, wages, and deflation

Some economic contracts are *indexed for inflation*, like Social Security retirement benefits and Treasury Inflation-Protected Securities (TIPS). By law, the cash values of these contracts adjust along with the officially reported level of consumer price index (CPI) inflation, keeping their real or inflation-adjusted values unaffected. By contrast, relatively few wage contracts are formally indexed for inflation. Unions and firms may bargain for or grant cost-of-living adjustments, and sometimes they are included in long-term contracts, but this is uncommon.

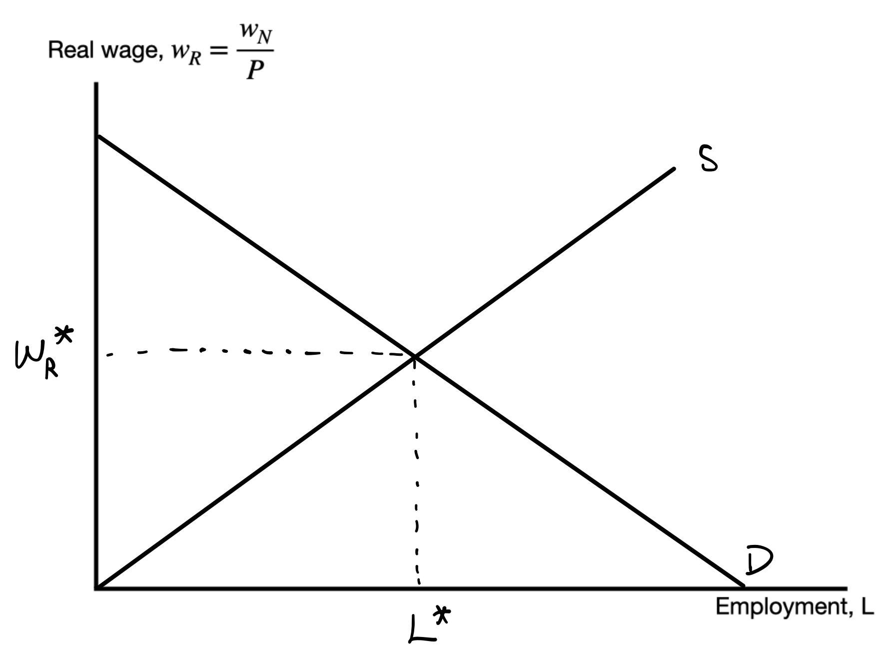
When inflation is low and predictable, this seems like an issue of negligible importance. But what if inflation unexpectedly becomes very low or very high?

Recall our model of the competitive national labor market, where labor demand is the downward sloping marginal product of labor, and the market sets the **real wage** w_R at the intersection of labor demand and an upward-sloping labor supply curve. But labor market agreements typically refer to the **nominal wage** w_N, a number like \$50 per hour or \$100,000 for 50 weeks of 40 hours. The relationship between these is a familiar one:

$$w_R = \frac{w_N}{P} \qquad (1.1)$$

where P is the price level.

(a) [2 points] On the axes below, draw a downward-sloping labor demand curve and an upward-sloping labor supply curve, and show the equilibrium real wage w_R^*.



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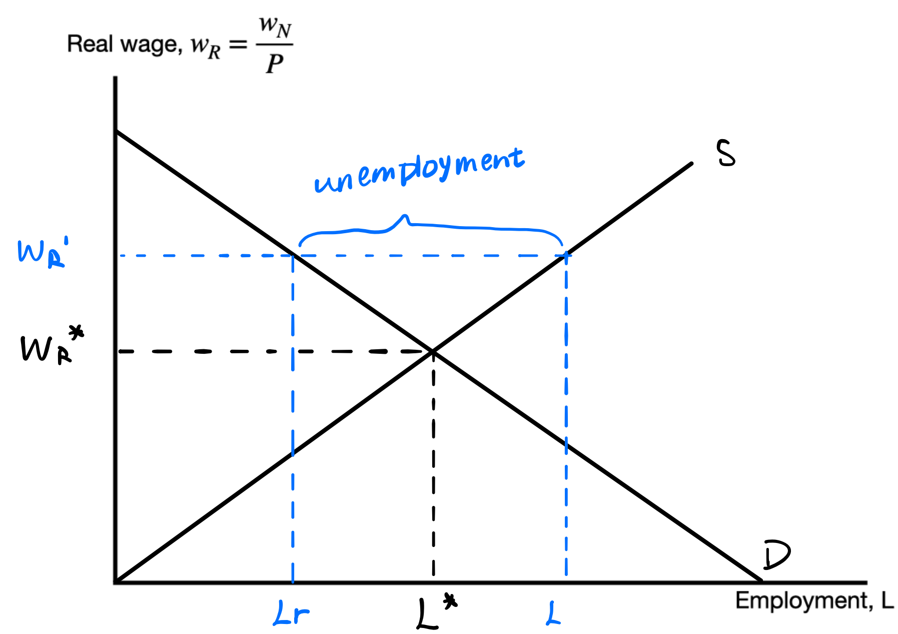
Suppose firms and workers in the labor market had to agree on a nominal wage that was implied by the real wage and expectations of the price level, P, so that $w_N = P \cdot w_R$ where the real wage is depicted in the graph you drew on the previous page.

Further, assume that firms cannot quickly renegotiate the nominal wage. They must either retain workers hired at the agreed nominal wage, or they must lay some or all workers off. Firms will always operate on the labor demand curve, although they might prefer to move to a different point along the curve if they could.

(b) [2 points] Suppose there is a large unanticipated *decrease in the price level* P, due either to a large price disinflation (π falls below its target) or deflation ($\pi < 0$). Given the constraints described above, what happens in the labor market when P unexpectedly falls by a lot? Describe any shifts in curves and any movements along curves.

when P unexpected falls, the nominal wage W_N is fixed. so the real wage $W_R = W_N/p$ rises above its equilibrium level W_R^*. No curves shift. Firms, which always operate on the labor demand curve, move up and to the left along D, reducing employment. Workers would like to supply more labor at the higher real wage, but firms only demand $L_r < L^*$.

(c) [2 points] Redraw your graph of the labor market and show the events of part (b). What, if anything, has happened to the real wage, to employment, and to unemployment?



Real wage increased, Employment decreased, Unemployment increased.

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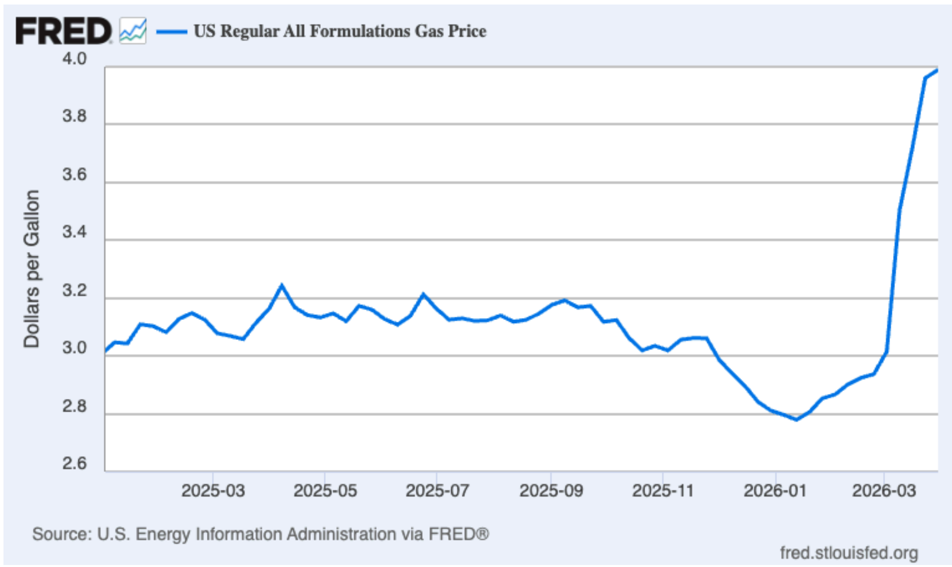
2. [14 points total] Shoe leather, tire rubber, and the smell of gasoline in Spring 2026.

The Federal Reserve's FRED database shows us [weekly average prices per gallon of regular gas] in the U.S., measured at 8:00 AM local time on Mondays and averaged across about 1,000 retail stations (source: [EIA]). The last data point shown is \$3.99 for March 30, 2026. On the last Monday in February, the average price was \$2.94.



(a) [2 points] Roughly how much in percentage terms had the average price of a gallon of regular gasoline risen during the month of March 2026?

$\% \Delta P = \frac{P_1 - P_0}{P_0} \times 100 = \frac{3.99 - 2.94}{2.94} \times 100 = \frac{1.05}{2.94} \times 100 \approx 35.7\%$ increase

(b) [2 points] Leaving aside the tiny "hook" visible at the extreme upper right of the graph, would you say that gas prices seem likely to be headed higher, the same, or lower?

I would say the prices are heading higher

(c) [2 points] Suppose you owned a car that runs on gasoline, and you noticed the fuel gauge was low. Normally, you don't even bother thinking about when to fill up the car with gas, because the time spent thinking about it is costly. Given the chart above, is there a cost associated with waiting to fill the tank? What might you do differently?

Yes, there is a real cost to waiting. If gas prices are rising, filling the tank today is strictly cheaper than filling it next week. To avoid this cost, we can just fill it up sooner and more fully.

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[Hyperinflation](#) is very rapid inflation, sometimes defined as a rate of 50% per month or more.

(d) [2 points] What annual rate of inflation corresponds to a sustained monthly rate of 50% inflation? (Hint: use the familiar relationship $x_t = x_0(1+g)^t$ where t indexes months.)

$(1+50\%)^{12} = 1.5^{12} \approx 129.75 \rightarrow g_{annual} = 129.75 - 1 = 128.75$

Annual inflation rate : 12875 %

(e) [2 points] If *all prices* were rising at an annual rate of 50%, would it make much sense to hold any domestic cash without spending it? Discuss. ("Domestic cash" can mean literally either paper or metal local currency, or it could mean a checking account or similar bank balance in that currency, on which no interest is earned.)

No, it does not make sense to hold domestic cash. The opportunity cost is too high. Rational agents will spend quickly or store value in non-cash items.

(f) [2 points] Thinking in terms of supply and demand for goods, what do the dynamics you described in parts (c) and (e) imply about market pressure(s) on prices — meaning shifts in either supply or demand or in both — absent any other changes?

part (c) : rising gas prices. part (e): high inflation.
Both behaviors in c and e shift demand curves to the right, which feeds back into higher prices.

(g) [2 points] Take a look at the complete time series of gas prices by clicking on the link above, which should show you data back to 1990. Given what you have seen in this problem set question thus far, and what you know about the state of the world in spring 2026, would you guess about the public's expectations regarding future gas prices?

The public likely expects gas prices to continue increasing in the near term.

3. [8 points total] MV = PY why why why why

Quantity Theory is very helpful in understanding historical macroeconomic fluctuations, and it remains useful today. Earlier in history, full-blown banking crises and financial panics were much more common than they are now, because modern practices in central banking had not yet been fully developed.

Recall that the quantity theory relationship is

$$M \cdot V = P \cdot Y^R \qquad (3.1)$$

where

- M is the money supply, which you should think of as the cash amount of currency and checking account balances (typically called $M1$)
- V is the velocity of money, the speed with which money moves through the economy during a time period, to purchase nominal GDP
- P is the price level. A common modern measure is the “GDP implicit price deflator”
- Y^R is real GDP, so the right-hand side is $P \cdot Y^R = Y^N$ or nominal GDP

Note that these modern time series behave in interesting ways, which reduces their usefulness in thinking about modern monetary policy.

Historical statistics for the money supply during the Great Depression were compiled by Milton Friedman and Anna Schwartz (1963). Gary Richardson (2013) outlines a key finding: “From the fall of 1930 through the winter of 1933, the money supply fell by nearly 30 percent.”

(a) [2 points] Assume that this development about M occurred without a change in V or Y^R. What must have happened to P? What does that mean about the *average annual inflation rate* π during this period? (Recall the familiar relationship $x_t = x_0(1 + g)^t$.)

When M fell 30% with roughly constant velocity, the price level fell significantly, which just the deflation. The larger the drop in M relative to the drop in Y, the more serve the deflation.

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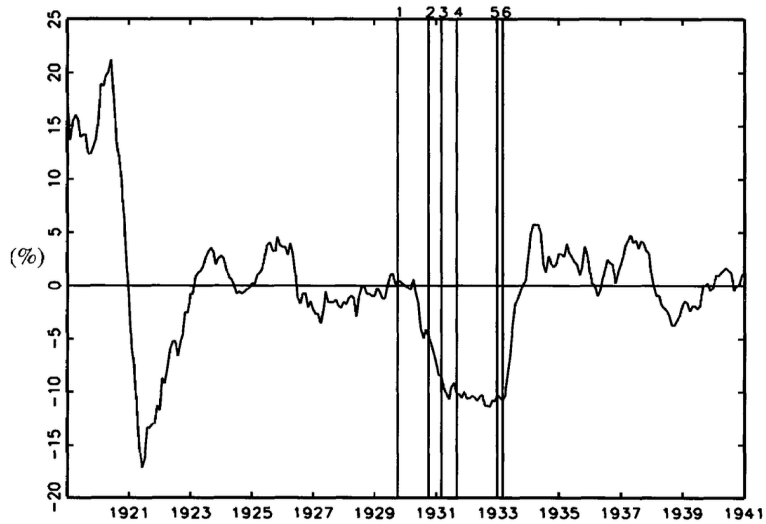
Figure 3 Monthly Changes in Consumer Price Inflation, 1919 to 1940



(b) [2 points] Recall Figure 3 from Cecchetti (1998) shown above and in class slides. Do the data shown there match what you reported in part (a)? Briefly discuss.

Yes. The Cecchetti figure shows the money supply and price level both falling during 1929-1933, which is exactly what the quantity theory predicts. The co-movement of M and P is consistent with MV=PY and constant V. The data validates the theoretical prediction that a sharp contraction in the money supply produced sustained deflation.

Recall that the Fisher equation shows the relationship between the nominal interest rate i, the real interest rate R, and the inflation rate π:

$$i = R + \pi \qquad (3.2)$$

(c) [2 points] Suppose that i remained roughly constant at $i = 0.02$. Given what you see in part (b) about π, what likely happened to R during 1930-1933? Did R increase, decrease, or stay the same, and at what level(s)?

$R = i - \pi \approx 10\% - 12\%$

(d) [2 points] Qualitatively, what is the likely effect on short-run output and employment of the change in R you described in part (c), given the slope of the IS curve?

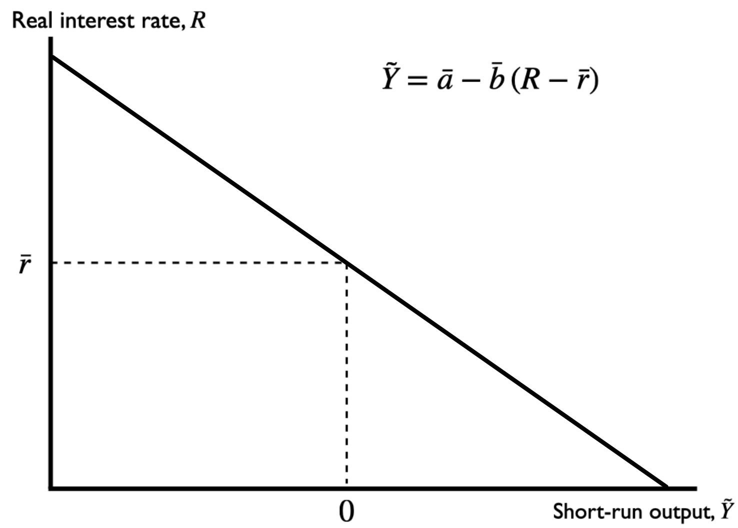
High $R \rightarrow (R - \bar{r})$ extremly larger than 0 $\rightarrow \tilde{Y} << 0$ in IS curve $\rightarrow$ serve output contraction

4. [12 points total] The IS curve, interest rates, and macroeconomic shocks.

Recall how the IS curve is a downward-sloping line in $(\tilde{Y}, R)$-space

$$\tilde{Y} = \bar{a} - \bar{b}\,(R - \bar{r}) \qquad (4.1)$$

(a) [2 points] Graph the IS curve on the axes below, being careful to position it correctly so that you are *depicting no temporary shocks* to aggregate demand.



Suppose that $\bar{b} = 0.4$ and $\bar{r} = 0.03$ so that

$$\tilde{Y} = \bar{a} - 0.4\,(R - 0.03) \qquad (4.2)$$

In each of the following scenarios, discuss what happens to the IS curve, if anything, and to short-run output $\tilde{Y}$, if anything. Begin each scenario by assuming that there is no temporary shock to aggregate demand $\bar{a} = 0$ and that monetary policy is neutral, $R = \bar{r}$. Both may change.

(b) [2 points] Suppose consumers become spooked by falling stock prices and temporarily reduce consumption spending by 0.1 percentage point of potential GDP.

A drop in consumer spending reduces aggregate demand directly → negative demand shock → $\bar{a}$ decreases → $\tilde{Y}$ is lower → IS curve shifts left

(c) [2 points] Suppose the Fed decides it must fight inflation (not shown here), and it raises the real interest rate to $R' = 0.04$.

Higher R → higher $R - \bar{r}$ → $\tilde{Y}$ falls
This is not a shift in the IS curve, it's a movement along the existing IS curve.

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(d) [2 points] Suppose Congress (the government) votes to go to war and passes a bill to raise government spending on military goods and services by $0.3 trillion or roughly 1 percent of GDP, and the President signs the bill.

For every level of R, $\tilde{Y}$ is now higher → IS curve shifts right

(e) [2 points] Suppose foreign economies temporarily increase their demand for U.S. exports by 0.3% of potential GDP.

same as (d), for every R, $\tilde{Y}$ rises → IS curve shifts right

(f) [2 points] Suppose the marginal product of capital $\bar{r}$ falls from $\bar{r} = 0.03$ to $\bar{r}' = 0.02$ but real interest rates remain at $R = 0.03$.

$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r}) \rightarrow \tilde{Y} = \bar{a} - \bar{b}R + \bar{b}\bar{r}$

$\Rightarrow$ if $\bar{r}$ falls, the term $\bar{b}\bar{r}$ decreases → for every R, $\tilde{Y}$ falls.

$\Rightarrow$ IS curve shifts left.